<u>Prometheus & Grafana Practical</u>

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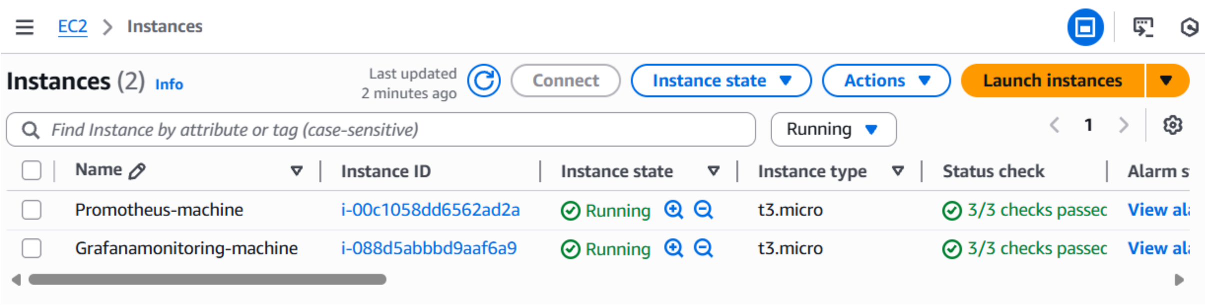
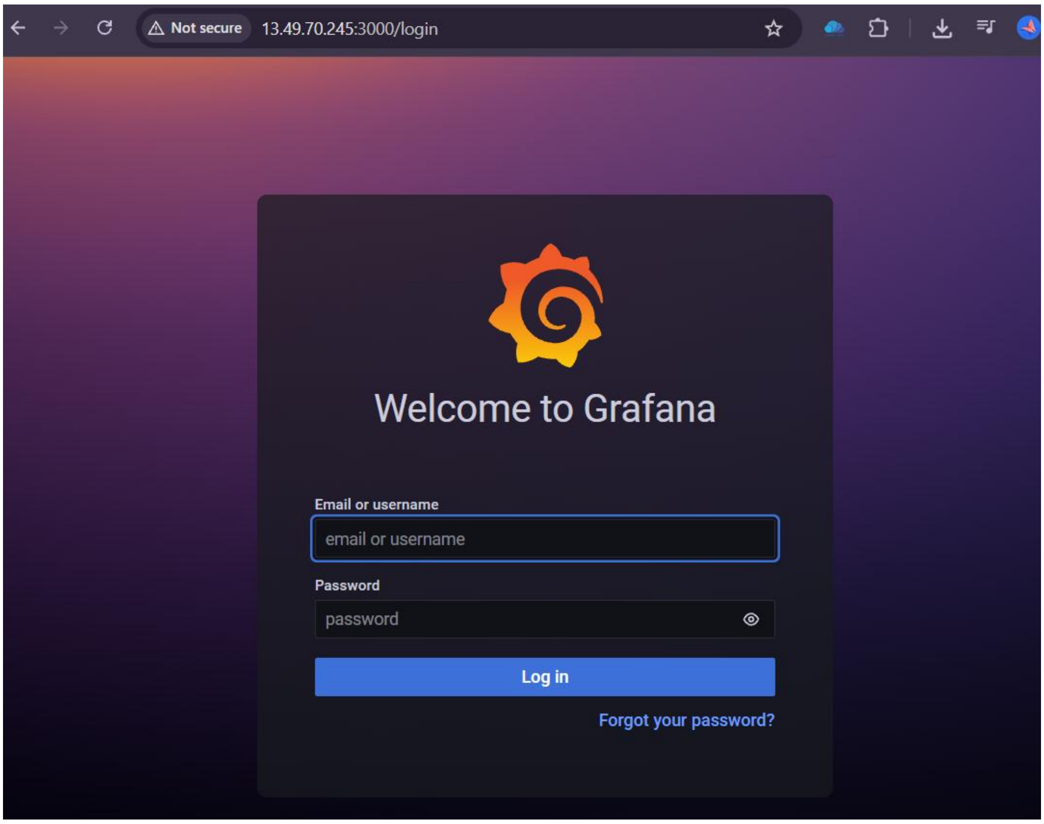
1. Docker Monitoring using Prometheus and Grafana

- Create two Ubuntu EC2 machines one for Prometheus and one for Grafana.
- Installation plan:
  - Grafana machine: Install Grafana.
  - Prometheus machine: Install Docker and Prometheus.



- Grafana-machine setup:
  - Grafana server runs on **port 3000**, so enable this port in the security group.
  - Access the Grafana webpage using the **public IP** of the machine and log in



- Prometheus-machine setup:
  - Install the Docker and Prometheus in the machine

```
root@ip-172-31-0-161:/home/ubuntu# ls
prometheus-2.34.0.linux-amd64  prometheus-2.34.0.linux-amd64.tar.gz
root@ip-172-31-0-161:/home/ubuntu# cd prometheus-2.34.0.linux-amd64
root@ip-172-31-0-161:/home/ubuntu/prometheus-2.34.0.linux-amd64# ls
LICENSE  NOTICE  console_libraries  consoles  prometheus  prometheus.yml  promtool
root@ip-172-31-0-161:/home/ubuntu/prometheus-2.34.0.linux-amd64#
```

  - Configure Docker to allow Prometheus to track it via **port 9323** by adding the required data to /etc/docker/daemon.json, then restart Docker.

```
{
        "metrics-addr" : "0.0.0.0:9323",
        "experimental" : true
}
~
~
~
```

  - Enable ports 9323 and 9090 in the security group of the Prometheus machine.

- To see Docker stats, access the Prometheus machine using **public IP:9323**.

```
Not secure   16.170.217.37:9323/metrics

# HELP builder_builds_failed_total Number of failed image builds
# TYPE builder_builds_failed_total counter
builder_builds_failed_total{reason="build_canceled"} 0
builder_builds_failed_total{reason="build_target_not_reachable_error"} 0
builder_builds_failed_total{reason="command_not_supported_error"} 0
builder_builds_failed_total{reason="dockerfile_empty_error"} 0
builder_builds_failed_total{reason="dockerfile_syntax_error"} 0
builder_builds_failed_total{reason="error_processing_commands_error"} 0
builder_builds_failed_total{reason="missing_onbuild_arguments_error"} 0
builder_builds_failed_total{reason="unknown_instruction_error"} 0
# HELP builder_builds_triggered_total Number of triggered image builds
# TYPE builder_builds_triggered_total counter
builder_builds_triggered_total 0
# HELP engine_daemon_container_actions_seconds The number of seconds it takes to process each container action
# TYPE engine_daemon_container_actions_seconds histogram
engine_daemon_container_actions_seconds_bucket{action="changes",le="0.005"} 1
engine_daemon_container_actions_seconds_bucket{action="changes",le="0.01"} 1
engine_daemon_container_actions_seconds_bucket{action="changes",le="0.025"} 1
engine_daemon_container_actions_seconds_bucket{action="changes",le="0.05"} 1
engine_daemon_container_actions_seconds_bucket{action="changes",le="0.1"} 1
```

- Update prometheus.yml by adding *- job_name: "docker"* to fetch Docker metrics.

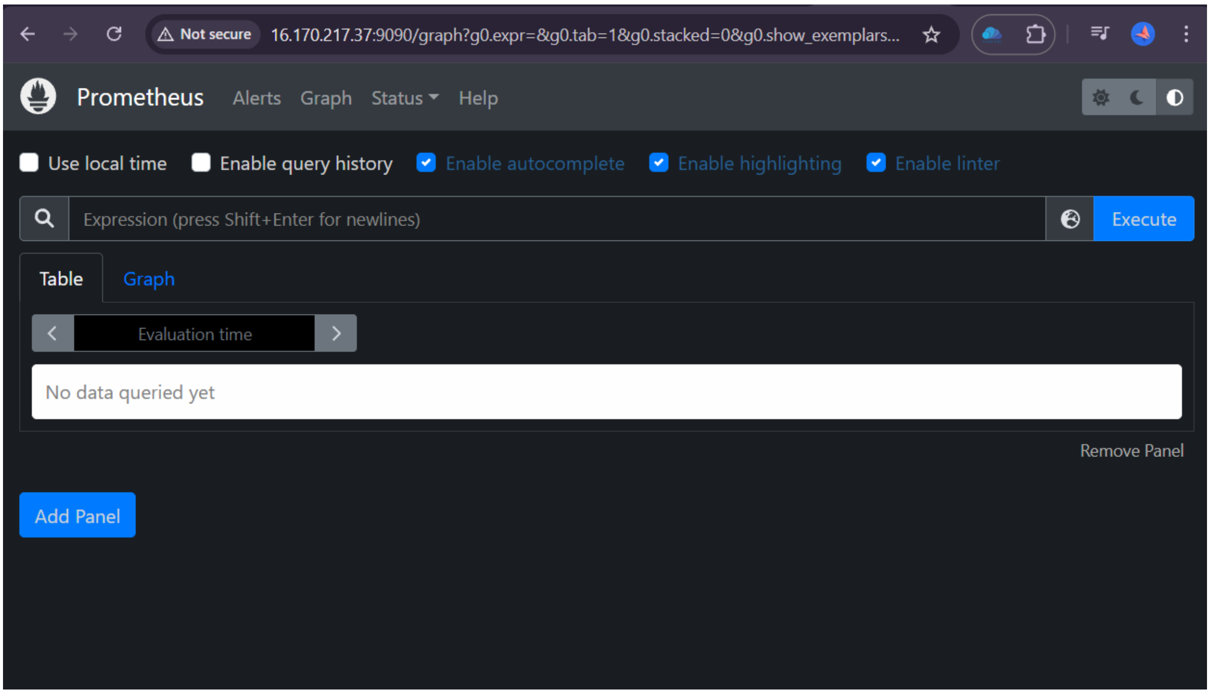
```
root@ip-172-31-0-161:/home/ubuntu/prometheus-2.34.0.linux-amd64# ls
LICENSE  NOTICE  console_libraries  consoles  prometheus  prometheus.yml  promtool
root@ip-172-31-0-161:/home/ubuntu/prometheus-2.34.0.linux-amd64# vi prometheus.yml
root@ip-172-31-0-161:/home/ubuntu/prometheus-2.34.0.linux-amd64#
```

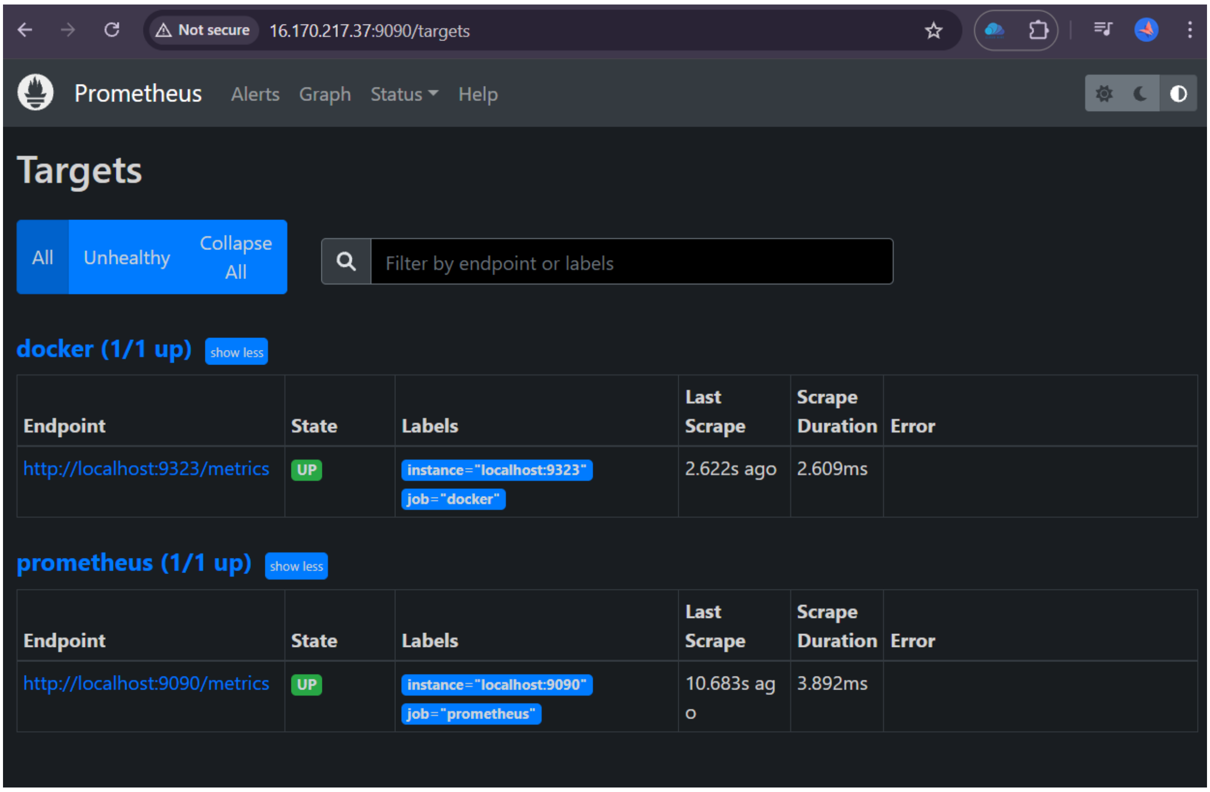
```
        - targets:
          # - alertmanager:9093

# Load rules once and periodically evaluate them according to the global 'evaluation_interval'.
rule_files:
  # - "first_rules.yml"
  # - "second_rules.yml"

# A scrape configuration containing exactly one endpoint to scrape:
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label `job=<job_name>` to any timeseries scraped from this config.
  - job_name: "prometheus"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9090"]
  - job_name: "docker"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9323"]
                                                                                35,18
```

- Start Prometheus by running *./prometheus*

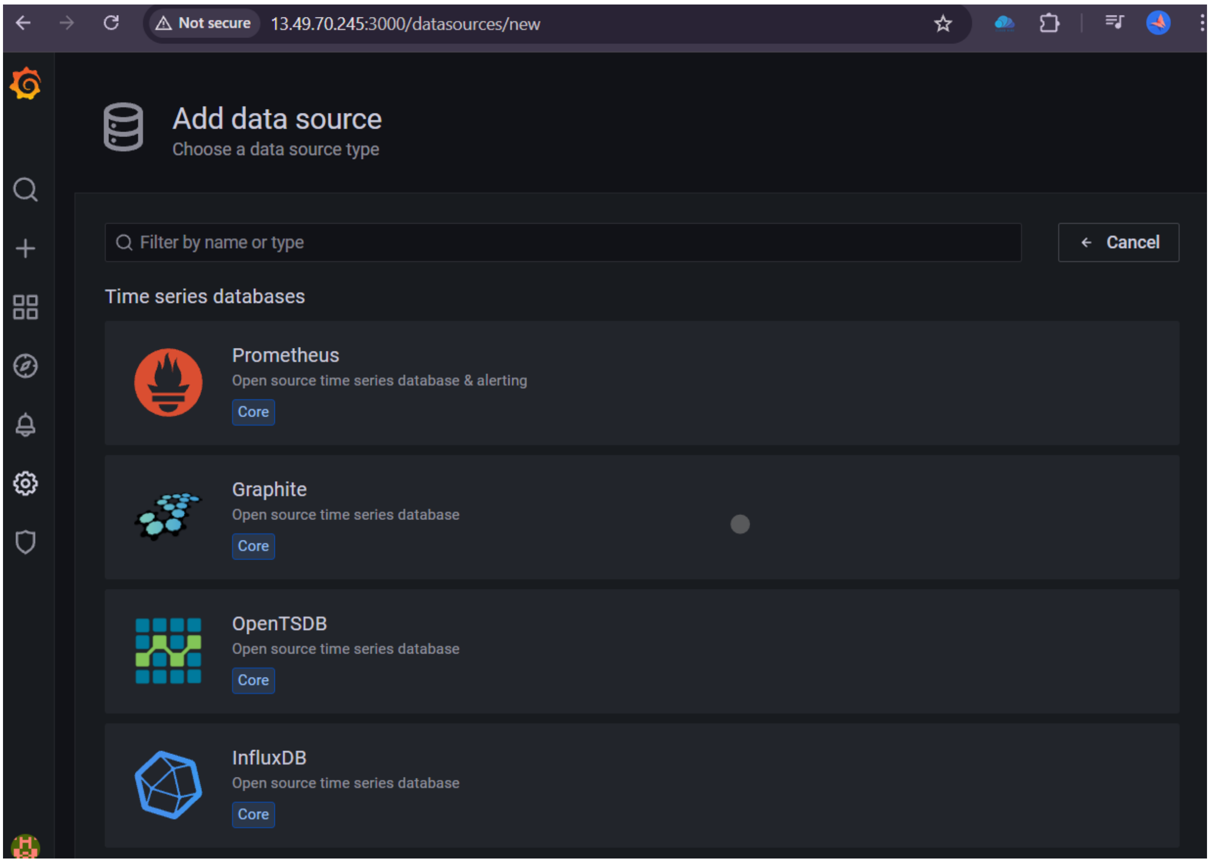
```
root@ip-172-31-0-161:/home/ubuntu/prometheus-2.34.0.linux-amd64# ls
LICENSE  NOTICE  console_libraries  consoles  prometheus  prometheus.yml  promtool
root@ip-172-31-0-161:/home/ubuntu/prometheus-2.34.0.linux-amd64# vi prometheus.yml
root@ip-172-31-0-161:/home/ubuntu/prometheus-2.34.0.linux-amd64# ./prometheus
ts=2026-02-12T09:49:17.164Z caller=main.go:479 level=info msg="No time or size retention was set so using th
e default time retention" duration=15d
ts=2026-02-12T09:49:17.164Z caller=main.go:516 level=info msg="Starting Prometheus" version="(version=2.34.0
, branch=HEAD, revision=881111fec4332c33094a6fb2680c71fffc427275)"
ts=2026-02-12T09:49:17.164Z caller=main.go:521 level=info build_context="(go=go1.17.8, user=root@121ad7ea548
7, date=20220315-15:18:00)"
ts=2026-02-12T09:49:17.164Z caller=main.go:522 level=info host_details="(Linux 6.14.0-1018-aws #18~24.04.1-U
buntu SMP Mon Nov 24 19:46:27 UTC 2025 x86_64 ip-172-31-0-161 (none))"
ts=2026-02-12T09:49:17.164Z caller=main.go:523 level=info fd_limits="(soft=1024, hard=1048576)"
ts=2026-02-12T09:49:17.164Z caller=main.go:524 level=info vm_limits="(soft=unlimited, hard=unlimited)"
ts=2026-02-12T09:49:17.168Z caller=web.go:540 level=info component=web msg="Start listening for connections"
 address=0.0.0.0:9090
ts=2026-02-12T09:49:17.168Z caller=main.go:937 level=info msg="Starting TSDB ..."
```

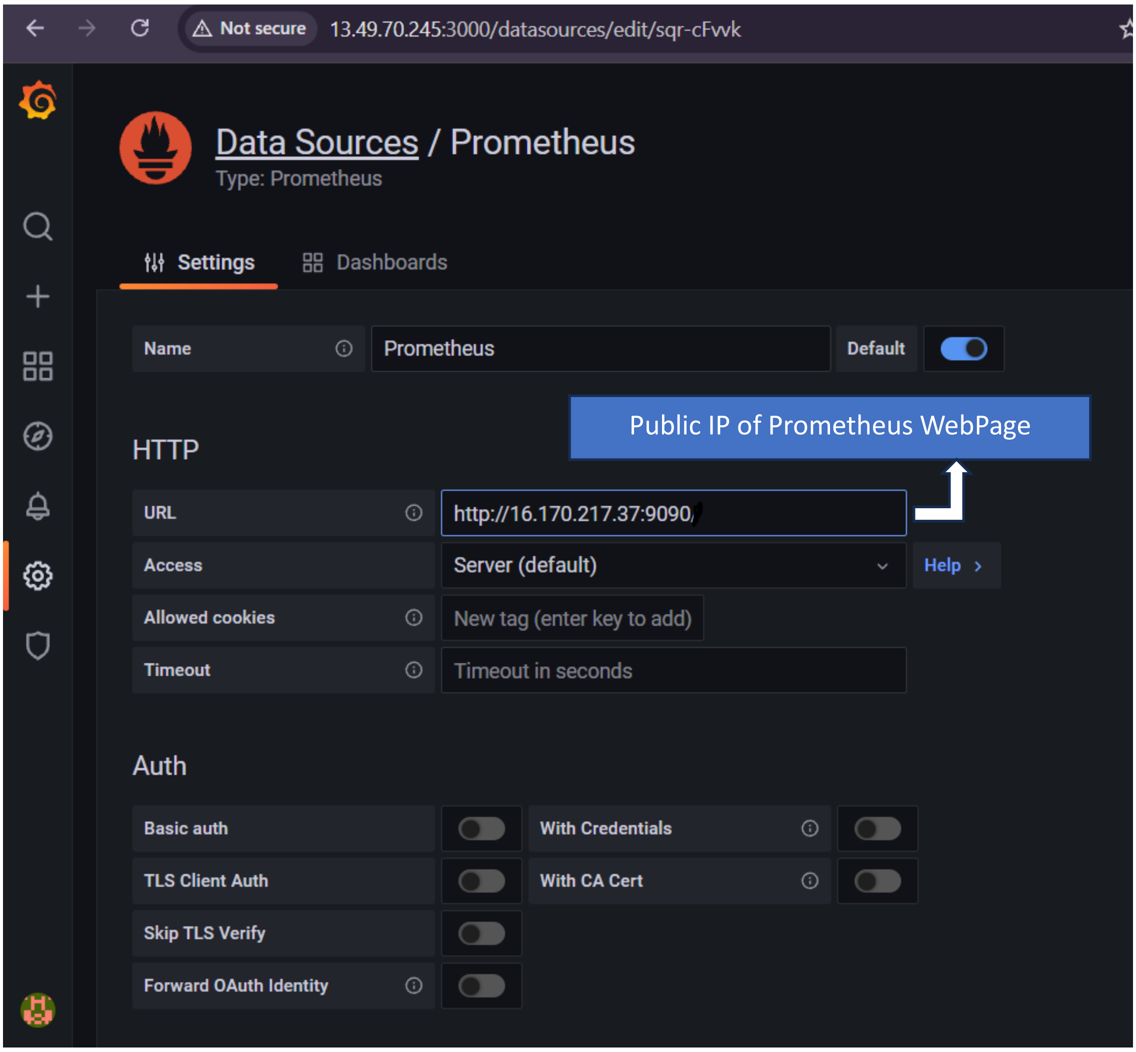
- Access the Prometheus webpage using public IP:9090



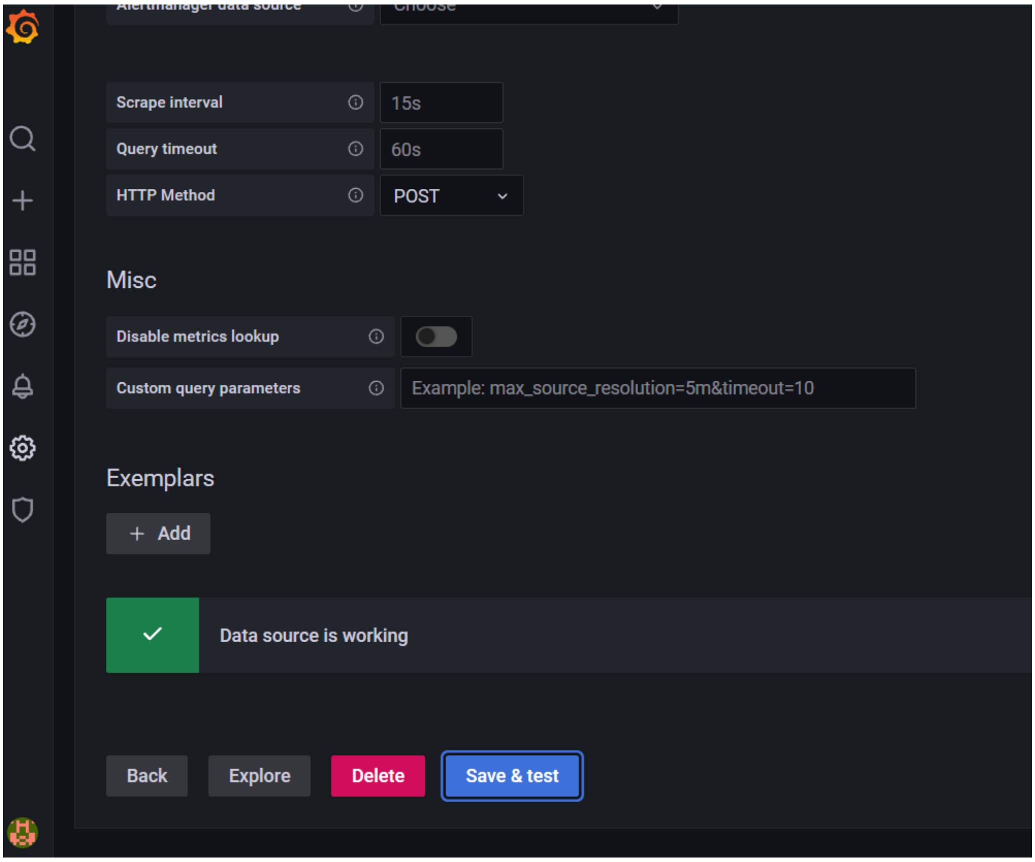


- Connect Grafana to Prometheus:
  - In Grafana, add Prometheus as a data source so that Grafana can visualize the metrics collected by Prometheus.

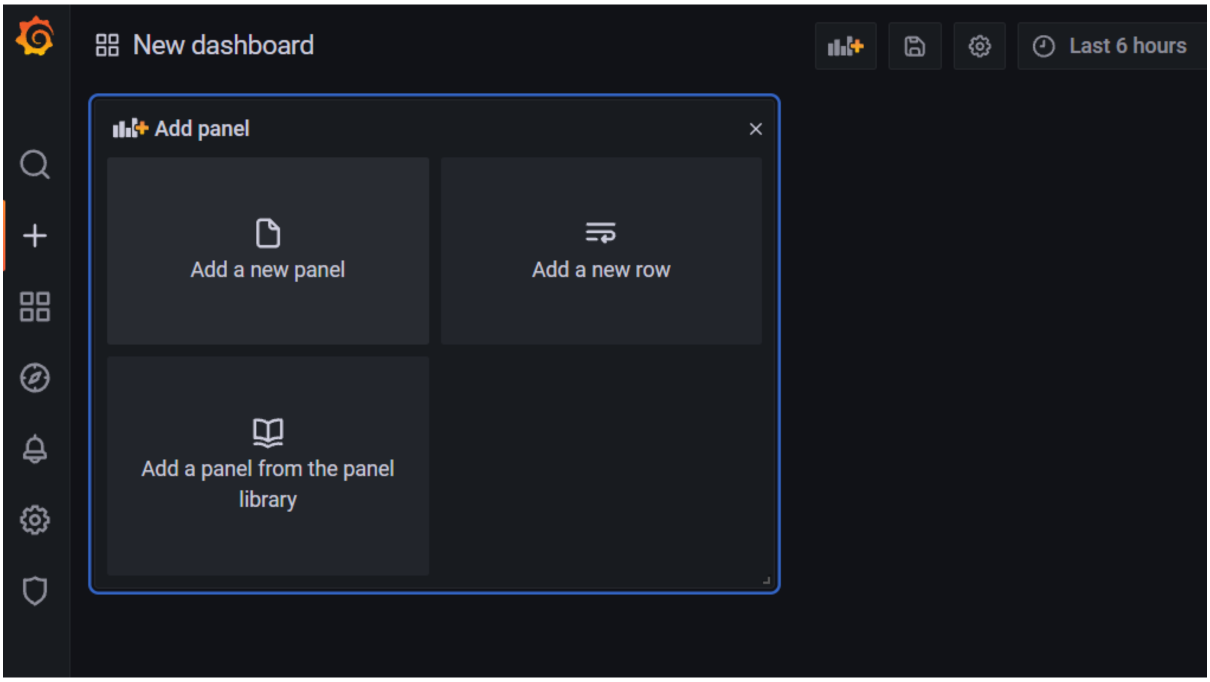


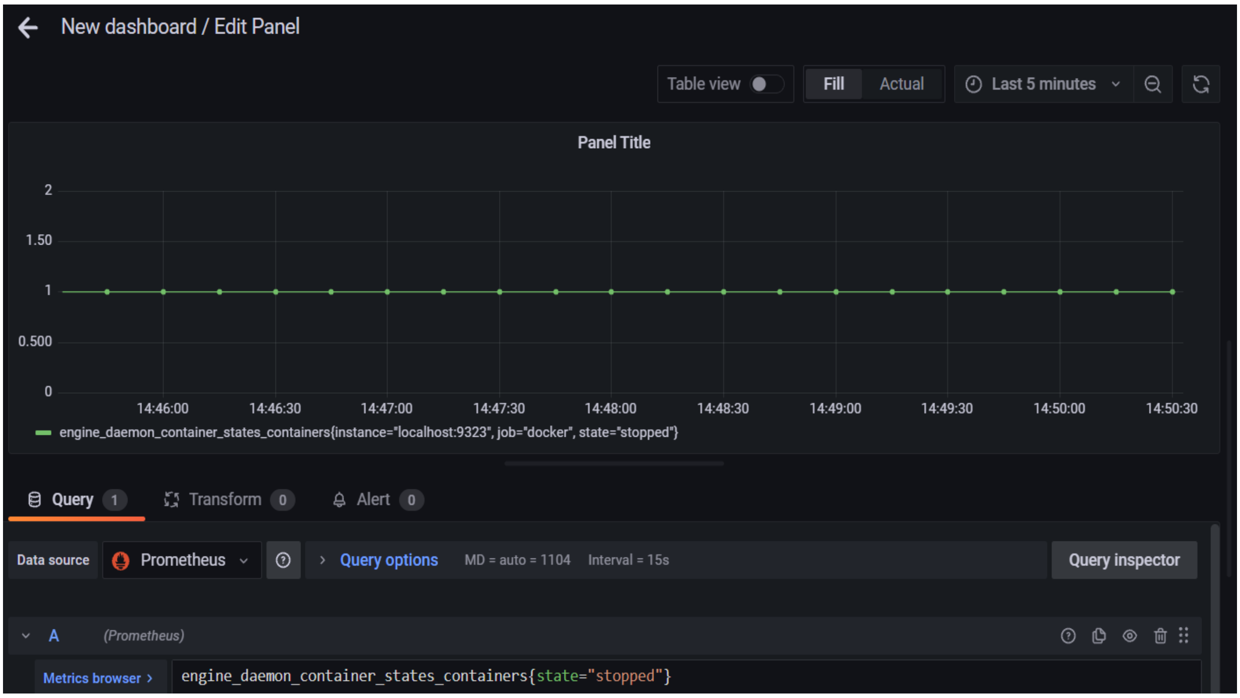


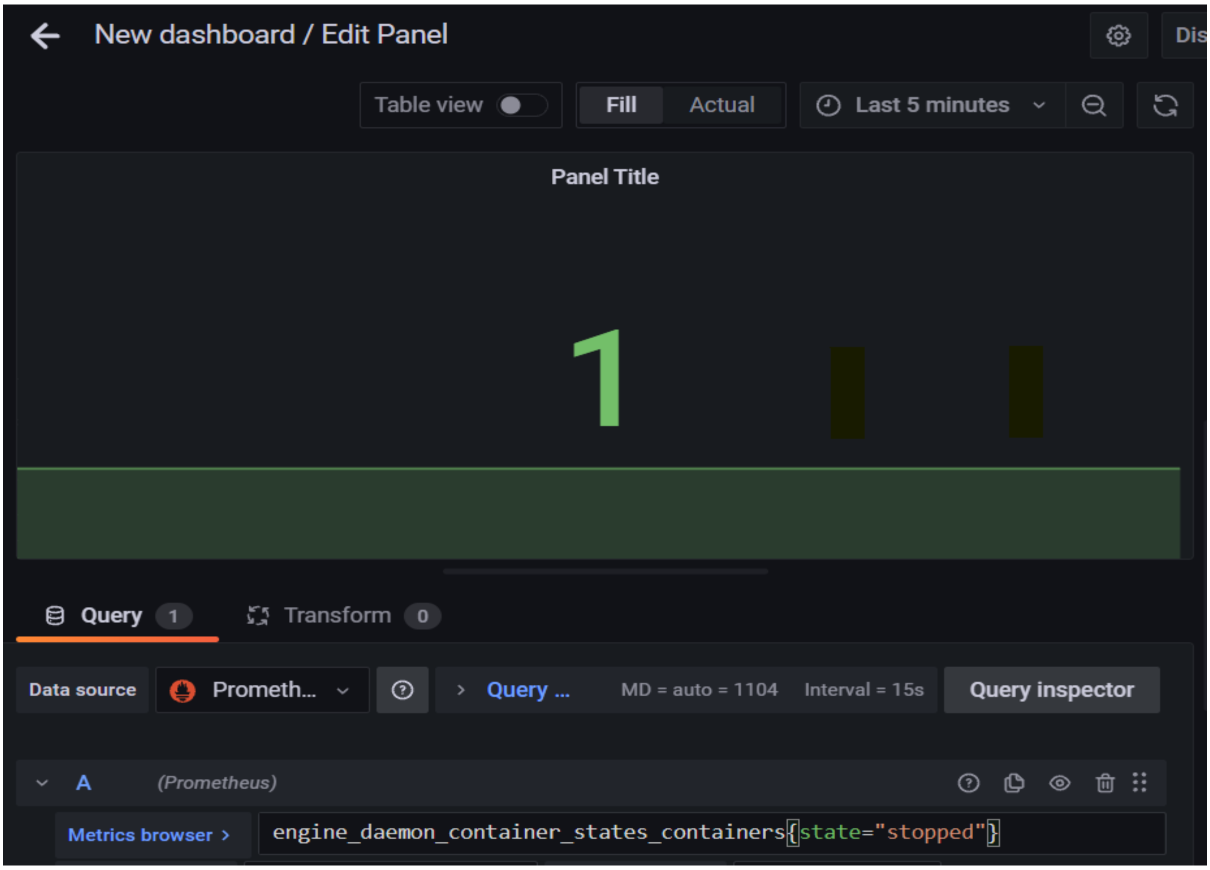
Not secure
13.49.70.245:3000/datasources/edit/sqr-cFvvk
Data Sources / Prometheus
Type: Prometheus
Settings
Dashboards
Name
Prometheus
Default
Public IP of Prometheus WebPage
HTTP
URL
http://16.170.217.37:9090/
Access
Server (default)
Help
Allowed cookies
New tag (enter key to add)
Timeout
Timeout in seconds
Auth
Basic auth
With Credentials
TLS Client Auth
With CA Cert
Skip TLS Verify
Forward OAuth Identity

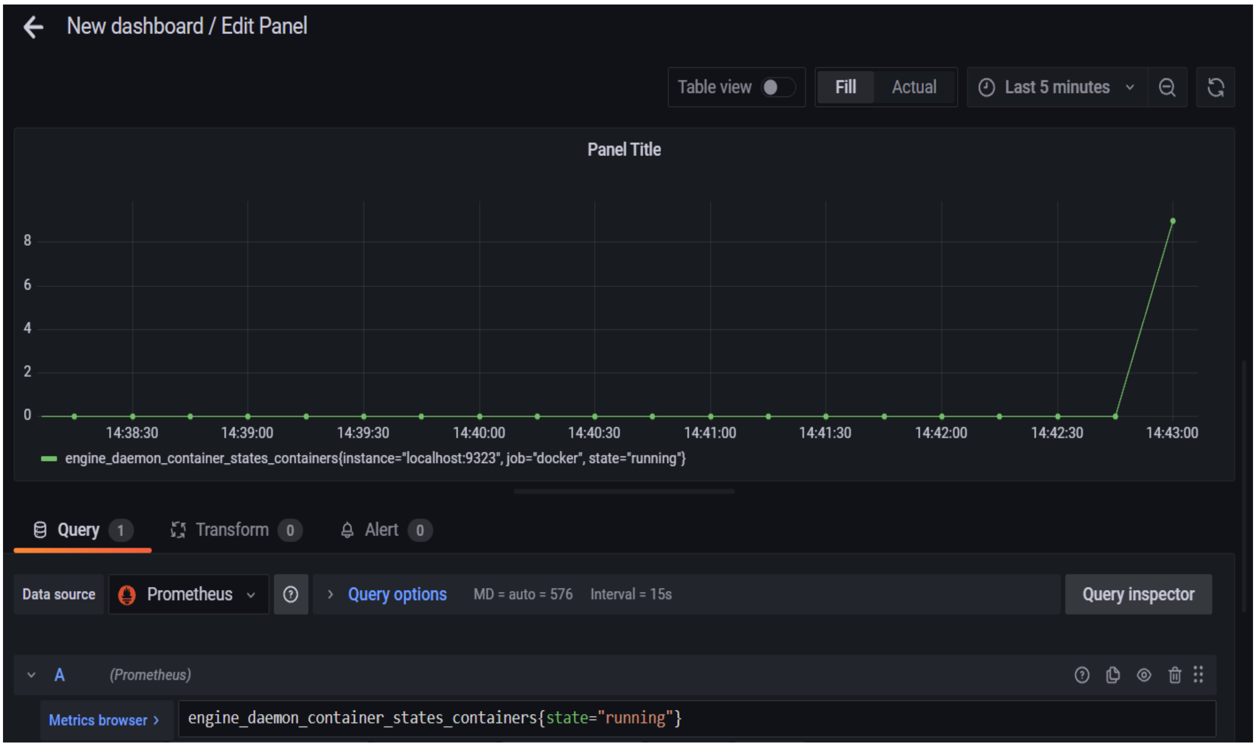


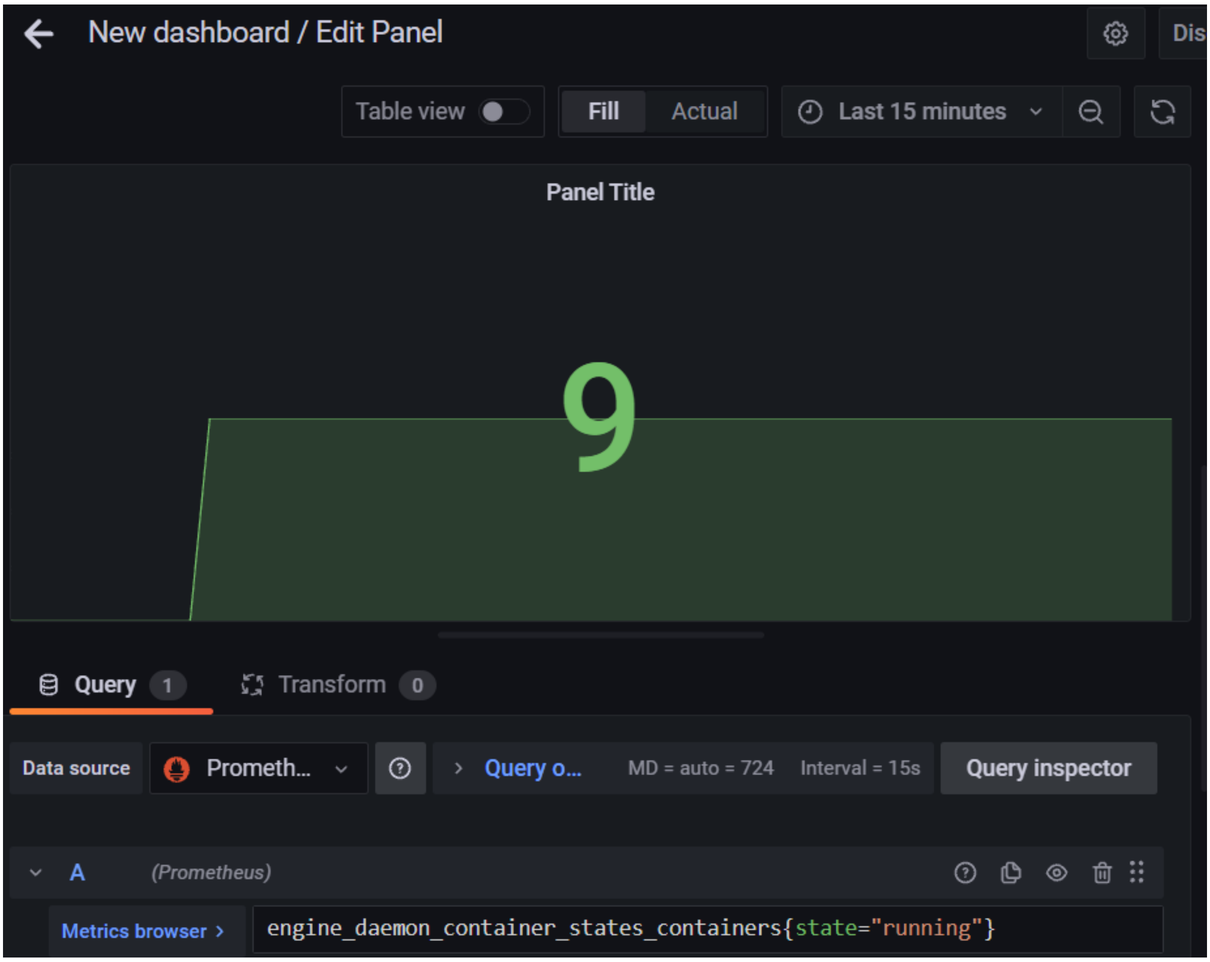
Scrape interval
15s
Query timeout
60s
HTTP Method
POST
Misc
Disable metrics lookup
Custom query parameters
Example: max_source_resolution=5m&timeout=10
Exemplars
Add
Data source is working
Back
Explore
Delete
Save & test


New dashboard
Last 6 hours
Add panel
Add a new panel
Add a new row
Add a panel from the panel library


New dashboard / Edit Panel
Table view
Fill
Actual
Last 5 minutes
Panel Title
engine_daemon_container_states_containers{instance="localhost:9323", job="docker", state="stopped"}
Query 1
Transform 0
Alert 0
Data source
Prometheus
Query options
MD = auto = 1104
Interval = 15s
Query inspector
A
(Prometheus)
Metrics browser
engine_daemon_container_states_containers{state="stopped"}


New dashboard / Edit Panel
Dis
Table view
Fill
Actual
Last 5 minutes
Panel Title
1
Query 1
Transform 0
Data source
Prometh...
Query ...
MD = auto = 1104
Interval = 15s
Query inspector
A
(Prometheus)
Metrics browser >
engine_daemon_container_states_containers{state="stopped"}


New dashboard / Edit Panel
Table view
Fill
Actual
Last 5 minutes
Panel Title
8
6
4
2
0
14:38:30
14:39:00
14:39:30
14:40:00
14:40:30
14:41:00
14:41:30
14:42:00
14:42:30
14:43:00
engine_daemon_container_states_containers{instance="localhost:9323", job="docker", state="running"}
Query 1
Transform 0
Alert 0
Data source
Prometheus
Query options
MD = auto = 576
Interval = 15s
Query inspector
A
(Prometheus)
Metrics browser >
engine_daemon_container_states_containers{state="running"}



New dashboard / Edit Panel
Dis
Table view
Fill
Actual
Last 15 minutes
Panel Title
9
Query 1
Transform 0
Data source
Prometh...
Query o...
MD = auto = 724
Interval = 15s
Query inspector
A
(Prometheus)
Metrics browser >
engine_daemon_container_states_containers{state="running"}